

MANTUO ORIGINAL FOUNDATION EDITION - ESAT - V0.1.0

ESAT Mathematics Starting Review Notes

ESAT 数学起点复习笔记

A curriculum-aware starting review for Mathematics 1 and Mathematics 2.

按 Mathematics 1 与 Mathematics 2 的真实知识单元整理课程差距、无计算器方法和第一轮复习动作。

The logo consists of the letters 'MT' in a bold, white, sans-serif font, centered within a solid purple square.

MANTUO ADMISSION TEST LIBRARY

不再为升学考试而焦虑

Mantou original foundation edition - independently authored for ESAT preparation

How to Use This Pack

如何使用这份笔记

IN THIS EDITION	This revision covers the exam map, three curriculum bridges, 15 top-level mathematics units, six methods, two original worked examples and six active-recall checks. 本版包含考试地图、A-Level/IAL、IB、AP 课程桥接、Mathematics 1 的 7 个一级知识单元、Mathematics 2 的 8 个一级知识单元、6 个解题方法、2 道原创逐步例题和 6 组主动回忆。
GO DEEPER	After this mathematics foundation, continue with the ESAT Science Foundation Notes, module-specific Advanced Strategy Notes and timed online papers, using real attempt evidence to choose the next training cycle. 完成这份数学基础版后，可继续使用 ESAT 理科基础笔记、按模块展开的深度策略笔记，以及在线限时试卷，用真实作答证据决定下一轮训练。
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FACTS FIRST

Exam Map

考试地图

PER MODULE	27 multiple-choice questions
每个模块	27 道选择题
TIME	40 minutes, separately timed
模块时间	40 分钟，独立计时
TOOLS	Calculators are not permitted
工具	不可使用计算器
SCORING	One mark per correct answer; no negative marking
计分	答对 1 分，答错不倒扣
SHARED FOUNDATION	All modules assume Mathematics 1 knowledge
共同基础	所有模块都默认 Mathematics 1 知识

A-LEVEL - IB - AP

Curriculum Bridge

课程衔接：具体缺什么

A curriculum can indicate likely exposure, not mastery. Confirm every claim against the student profile and timed evidence.

课程范围只说明可能学过什么，不等于已经掌握；请结合本人档案和真实练习。

A-Level / International A-Level Mathematics	STRONG START 大部分知识通常已出现
LIKELY TRANSFER	- Algebra, functions, trigonometry, exponentials, logarithms and introductory calculus 代数、函数、三角、指数对数与基础微积分 - Core ideas in sequences, coordinate geometry, statistics and probability 数列、坐标几何、统计与概率的核心概念
CHECK ONE BY ONE	- Fractions, surds, unit conversion and order-of-magnitude estimation without a calculator 无计算器下的分数、根式、单位换算和数量级估算 - Whether conditional probability, graph transformations and coordinate geometry of circles can be retrieved under time pressure 条件概率、函数图像变换与圆的坐标几何是否能在限时内直接调用
FIRST ACTION	Check all 15 knowledge units, then use one 40-minute practice set to distinguish retrieval gaps from pacing problems. 先逐项检查 15 个知识单元，再用一套 40 分钟练习判断问题属于知识调用还是节奏。

IB Mathematics AA / AI	PARTIAL 课程路径决定覆盖程度
LIKELY TRANSFER	- AA HL usually provides a strong starting point in functions, trigonometry and calculus AA HL 通常对函数、三角与微积分起点较强 - AI courses usually provide greater familiarity with data, statistics and modelling language AI 课程通常对数据、统计与建模表述更熟悉
CHECK ONE BY ONE	- Exact algebra, surds, non-calculator arithmetic and coordinate geometry 精确代数、根式、无计算器运算与坐标几何 - Sequences, radians, logarithmic identities and graph reasoning without a graphical calculator 数列、弧度、对数恒等式以及不依赖图形计算器的图像判断
FIRST ACTION	Do not rely on the AA or AI label alone. Check your completed units and confirm that you can work without a formula booklet or graphical calculator. 不要只按 AA/AI 标签判断；回到本人完成的单元，逐项确认能否脱离公式表与图形计算器。

AP Precalculus / Calculus / Statistics	PARTIAL 需要组合课程证据
LIKELY TRANSFER	- Precalculus and Calculus may cover parts of functions, trigonometry and calculus Precalculus 与 Calculus 可覆盖部分函数、三角和微积分 - Statistics may cover parts of data representation and probability Statistics 可覆盖部分数据表示和概率

CHECK ONE BY ONE	- SI units, compound units, ratios and exact-value techniques common in UK curricula SI 单位、复合单位、比例与英国课程中的精确数值技巧 - Sequences, coordinate geometry of circles, conditional probability, exponentials, logarithms and non-calculator trigonometry 数列、圆的坐标几何、条件概率、指数对数和无计算器三角
FIRST ACTION	Treat completion of Calculus separately from full mathematics coverage. Close unsupported Mathematics 1 gaps before checking Mathematics 2. 把修过 Calculus 与数学全覆盖分开；先补没有课程证据的 Mathematics 1 单元，再检查 Mathematics 2。

MODULE 01 - ESAT

Mathematics 1 - Shared Mathematical Foundation

Mathematics 1: 共同数学基础

Every ESAT module assumes immediate access to Mathematics 1. The aim is not to memorise more formulae, but to translate units, quantities, ratios, algebra and data displays accurately without a calculator.

所有 ESAT 模块都默认能够调用 Mathematics 1。重点不是背更多公式，而是在无计算器条件下准确翻译单位、数量、比例、代数与图表。

LEARNING OUTCOMES	- Audit curriculum evidence against M1-M7 按 M1-M7 逐项检查课程证据 - Resolve units and relationships before calculating 先处理单位和关系，再进行运算 - Use estimates, bounds and answer options to test a result 用估算、边界与选项检查结果
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KNOWLEDGE MAP

Check Every Knowledge Unit

逐项核对知识单元

M1	Units 单位与复合单位
M2	Number 数与精确计算
M3	Ratio and proportion 比、比例与变化率
M4	Algebra 代数、方程与函数

M5	Geometry 几何、三角与向量
M6	Statistics 统计与数据表示
M7	Probability 概率与条件概率

QUESTION METHODS

What to Notice, How to Start, What to Check

看到什么，怎样开始，如何检查

01	NORMALISE UNITS 先统一单位
TRIGGER	The question mixes minutes and seconds, litres and millilitres, or other compound units. 题目同时出现分钟、秒、升、毫升或复合单位。
ACTION	State the target unit on the first line. Make only the necessary conversions before substituting into the quantitative relationship. 在第一行写出目标单位；只做必要换算，再代入数量关系。
CHECK	The unit, order of magnitude and direction of change must all be plausible. 答案的单位、数量级和方向必须同时合理。
02	TRANSLATE RELATIONSHIPS 把文字变成关系
TRIGGER	The wording includes 'per', 'increases to', 'decreases by' or 'is proportional to'. 题目使用“每”“增加到”“减少了”“成比例”等表达。
ACTION	Define the reference quantity and change factor before combining any numbers. 先定义基准量与变化因子，不要直接抓住页面上的数字相乘。
CHECK	Substitute the result back into the sentence and confirm that the reference quantity was not reversed. 把结果代回原句，确认比较基准没有被交换。
03	ESTIMATE FIRST 先估算再精算
TRIGGER	The options are widely separated, or exact arithmetic is likely to create sign or decimal errors. 选项相距较远，或精确运算容易产生符号/小数错误。

ACTION	Fix the sign, scale and plausible interval first, then perform the minimum exact calculation required. 先确定正负、数量级和合理区间，再完成最少的精确运算。
CHECK	If the exact value sits far outside the estimate, check units and the direction of the ratio first. 若精确答案离估算区间很远，优先检查单位或比例方向。

ORIGINAL WORKED EXAMPLE

Mantou original - flow rate and time units

满托原创例题：流量与时间单位

PROBLEM	Water flows into a container at 1.8 litres per minute for 35 seconds. How many millilitres enter? 水以每分钟 1.8 升的恒定流量流入容器，持续 35 秒。流入多少毫升？
Set the target unit 确定目标单位	The answer is required in millilitres. Convert the time to minutes while keeping the flow rate in litres per minute. 答案需要毫升；先把时间写成分钟，流量单位暂时保持升/分钟。 35 seconds equals 35 over 60 minutes
Apply the flow-rate relationship 应用流量关系	Volume equals flow rate multiplied by time. 体积等于流量乘时间。 1.8 times 35 over 60 equals 1.05 litres
Convert to the requested unit 换成题目单位	One litre is 1000 millilitres. 一升等于一千毫升。 1.05 litres equals 1050 millilitres
CONCLUSION	The volume entering the container is 1050 mL. 流入 1050 mL。
COMMON TRAP	Multiplying 35 seconds directly by a per-minute rate produces inconsistent dimensions. Make the units compatible before calculating. 把 35 秒直接与“每分钟”相乘，会得到量纲错误的结果；单位一致必须发生在运算前。

Close the Notes Before You Answer

合上笔记再回答

01	What should your first line show when compound units appear? 看到复合单位时，第一行应该写什么？
ANSWER	State the target unit and convert the relevant quantities to compatible units. 写出目标单位，并把所有相关量换到相容单位。

02	Why should estimation come before exact calculation? 为什么估算不是精确运算后的附加步骤？
ANSWER	It fixes sign, scale and range, exposing unit or ratio errors before and after calculation. 它先锁定符号、数量级和区间，可以在精算前后捕捉比例或单位错误。

03	What multipliers represent an increase by 20% and an increase to 120%? “增加了 20%”与“增加到 120%”的乘数分别是什么？
ANSWER	Both correspond to multiplying by 1.2 here, but the original reference quantity must be identified first. 两者在该例中都对应乘以 1.2，但必须先辨认原始基准量。

Mathematics 2 - Structure and Reasoning

Mathematics 2: 结构与推理

Mathematics 2 assumes the whole Mathematics 1 foundation and requires you to switch between algebra, graphs, sequences, trigonometry and calculus. Completing more curriculum content does not automatically produce better timed decisions.

Mathematics 2 需要 Mathematics 1 的全部基础，并进一步要求在代数、图像、数列、三角与微积分之间切换表示。更多课程内容不会自动转化为更好的限时判断。

LEARNING OUTCOMES

- Build a complete gap list across MM1-MM8

按 MM1-MM8 建立完整缺口清单

- Identify structure and constraints before choosing an algorithm

先识别结构与约束，再选择算法

- Use graphs, discriminants, bounds or options to reduce calculation

用图像、判别式、边界或选项减少运算

KNOWLEDGE MAP

Check Every Knowledge Unit

逐项核对知识单元

MM1	Algebra and functions 代数与函数
MM2	Sequences and series 数列与级数
MM3	Coordinate geometry 坐标几何与圆
MM4	Trigonometry 三角函数与弧度
MM5	Exponentials and logarithms 指数与对数
MM6	Differentiation 微分及其应用
MM7	Integration 积分、面积与微分方程

MM8	Graphs of functions 函数图像与变换
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QUESTION METHODS

What to Notice, How to Start, What to Check

看到什么，怎样开始，如何检查

01	IDENTIFY STRUCTURE 先找结构
TRIGGER	Expansion is possible, but the conditions suggest symmetry, factors, a discriminant or a graphical relationship. 题目可以展开计算，但条件暗示对称、因式、判别式或图像关系。
ACTION	Write down the smallest structure that determines the answer, such as the number of roots, an interval of monotonicity or an invariant. 写出决定答案的最小结构，例如根的个数、单调区间或不变量。
CHECK	If the solution requires a long expansion, ask whether the given structure has been ignored. 如果解法需要长串展开，先问是否忽略了题目给出的结构。
02	SWITCH REPRESENTATIONS 切换表示
TRIGGER	An algebraic form is hard to compare directly but can be graphed, completed, parameterised or interpreted as an area or gradient. 代数式难以直接比较，但可以画图、配方、参数化或解释为面积/斜率。
ACTION	Choose the representation that exposes the constraints most directly instead of persisting with the first form presented. 选择能最直接暴露约束的表示，而不是坚持第一眼看到的形式。
CHECK	After switching representation, confirm that the domain, signs and endpoints are still present. 转换后确认定义域、符号与端点没有丢失。
03	TEST THE OPTIONS 利用选项验证
TRIGGER	There are few answer options and substitution is shorter than a full solution. 选择题选项数量少，反向代入比完整求解更短。
ACTION	Eliminate with signs, ranges and special values, then perform the minimum check on the remaining options. 先用符号、范围和特殊值排除，再对剩余选项做最小验证。
CHECK	An option must satisfy every condition, not merely one equation. 选项法必须满足全部条件，不能只匹配其中一个等式。

Mantou original - determine a parameter from the number of roots

满托原创例题：用根的个数锁定参数

PROBLEM	The graph of $f(x)=x^2-4x+k$ has exactly one point in common with the x-axis. Find k. 函数 $f(x)=x^2-4x+k$ 的图像与 x 轴恰好有一个公共点。求 k。
Translate the graph into roots 把图像语言翻译为根	Exactly one common point means that the quadratic has a repeated root, so the discriminant is zero. 恰好一个公共点表示二次方程有一个重根，因此判别式为零。
Write the minimum condition 写出最小条件	For ax^2+bx+c, the discriminant is b^2-4ac. 对 ax^2+bx+c, 判别式是 b^2-4ac。 negative four squared minus four times one times k equals zero
Solve for the parameter and check 解参数并检查	This gives $k=4$. Then $f(x)=(x-2)^2$, which touches the x-axis only at $x=2$. 得到 $k=4$; 此时 $f(x)=(x-2)^2$, 只在 $x=2$ 接触 x 轴。 k equals 4 and f of x equals x minus 2 squared
CONCLUSION	$k=4$. $k=4$。
COMMON TRAP	Using only 'has a real root' gives $k \leq 4$. The question requires exactly one common point, so the discriminant must equal zero. 只说“有实根”会得到 $k \leq 4$; 题目要求恰好一个公共点，所以必须使用判别式等于零。

ACTIVE RECALL

Close the Notes Before You Answer

合上笔记再回答

01	What does exactly one x-intercept imply about the discriminant? 二次函数与 x 轴恰有一个交点，对判别式意味着什么？
ANSWER	The discriminant is zero, giving a repeated root. 判别式等于零，对应一个重根。
02	When should you switch from algebra to a graph? 何时应该从代数式切换到图像？
ANSWER	When roots, sign, intersections, monotonicity or transformations matter more directly than expansion. 当问题关心根的个数、符号、交点、单调或变换，而展开不能直接暴露这些关系时。

03

What is most often missed when testing answer options?

用选项反代时，最容易漏掉什么？

ANSWER

Checking only part of the conditions, or losing domain, endpoint or sign restrictions.

只验证部分条件，或忘记定义域、端点与符号限制。

Review Loop

把复习变成下一步行动

01	TAG THE UNIT 标记单元	Map each missed question to M1-M7 or MM1-MM8 before using a vague label such as 'careless'. 每道错题先对应到 M1-M7 或 MM1-MM8，不用“粗心”代替原因。
02	SEPARATE KNOWLEDGE FROM RETRIEVAL 区分知识与调用	Decide whether the content is unlearned, known but unavailable without prompting, or represented inefficiently under time pressure. 判断是没有学过、知道但无法无提示调用，还是时间压力下选择了低效表示。
03	RECORD THE SHORTEST VALID PATH 写下最短正确路径	Record only the relationship that determines the answer, the essential checks and the unproductive entry route to avoid. 只记录决定答案的关键关系、检查点和应当放弃的错误入口。
04	ASSIGN THE NEXT ACTION 安排下一动作	Send knowledge gaps to review, retrieval gaps to a short same-tag set, and pacing problems to a separately timed 40-minute module. 知识缺口进入复习；调用问题做同标签短组；节奏问题进入独立 40 分钟模块。

SOURCES

Version Boundary and Official Anchors

版本边界与官方依据

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ESAT Content Specification - October 2026 and January 2027

Used to verify the five-module structure, 27-question and 40-minute format, separate timing, calculator rule, scoring and the top-level Mathematics 1 and 2 scope.

核对五模块结构、27 题/40 分钟、独立计时、无计算器、计分规则和 Mathematics 1/2 的一级知识范围。

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Notes on Mathematics for TMUA and ESAT Mathematics 2

Used only to verify the MM1-MM8 scope and official terminology for Mathematics 2; all methods, examples and curriculum bridges on this site are independently authored.

只核对 Mathematics 2 的 MM1-MM8 范围与官方术语；本站方法、例题和课程桥接均为独立创作。

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